

An Explicit Formula for the Loday Assembly

Virgil Chan

How I Tortured My Advisor

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* Introduction

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The presented work deals with the Isomorphism Conjecture in Algebraic K-theory.

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The Loday assembly

$$\alpha_L : BG_+ \wedge K_R^{GW} \rightarrow K_{R[G]}^{GW}$$

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Classifying space of the discrete group G

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disjoint base-point

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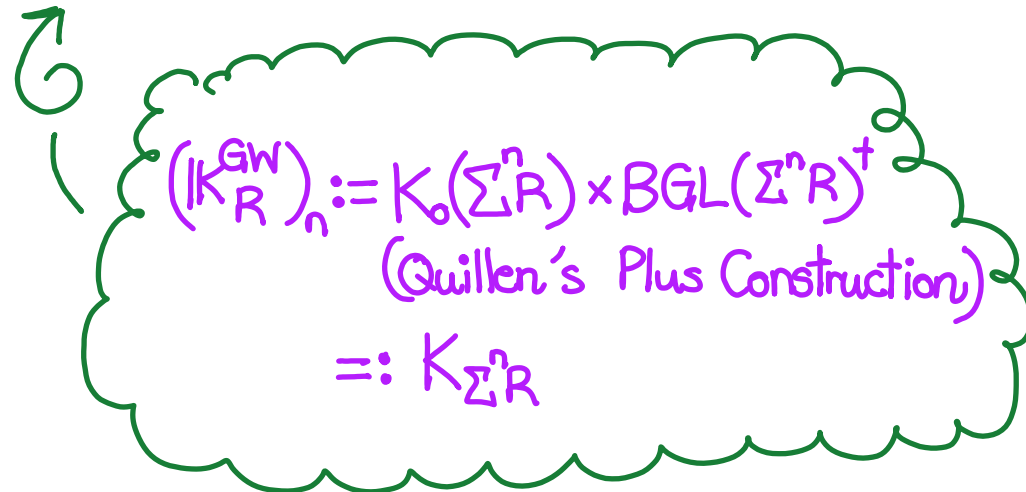
non-connective
Gersten-Wagoner
algebraic K-theory spectrum
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$(K_R^{GW})_n := K_0(\Sigma^n R) \times BGL(\Sigma^n R)^+$
(Quillen's Plus Construction)
 $=: K_{\Sigma^n R}$

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The Loday assembly

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Group ring

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{ All rings are unital,
associative }

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Homotopy groups

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Homotopy groups

RHS:

$$\pi_i(K_{R[G]}^{GW}) =: K_i(R[G]) \quad (\text{algebraic K-groups of } R[G])$$

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Homotopy groups

Difficult to compute!

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Homotopy groups

LHS: represents a generalised homology theory

So, the Loday assembly approximates homotopy by homology.

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$$\alpha_L : BG_+ \wedge K_R^{GW} \rightarrow K_{R[G]}^{GW}$$

Homotopy groups

LHS: represents a generalised homology theory

Have the Atiyah-Hirzebruch spectral sequence :

$$E_{p,q}^2 \cong H_p(BG; K_q(R)) \Rightarrow \pi_{p+q}(BG_+ \wedge K_R^{GW})$$

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$$\alpha_L : BG_+ \wedge K_R^{GW} \rightarrow K_{R[G]}^{GW}$$

Homotopy groups

LHS: represents a generalised homology theory

Therefore, we think of the LHS as more computable.

* Isomorphism Conjecture

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If R is a regular ring and G is a torsion-free group, then

$$\pi_i(\alpha_L) \otimes \text{id}_{\mathbb{Q}} : \pi_i(BG_+ \wedge K_R^{GW}) \otimes \mathbb{Q} \longrightarrow K_i(R[G]) \otimes \mathbb{Q}$$

is injective for all $i \in \mathbb{Z}$.



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$K_i(R) = 0$ for $i < 0$.
So, the spectrum K_R^{GW}
is connective

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Thm ① True if $H_s(BG; \mathbb{Z})$ is finitely-generated for all s (Bökstedt - Hsiang - Madsen 1993).

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Thm ② True integrally if G has finite decomposition complexity and has a finite model for EG (Ramras-Tessera-Yu 2014).



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Thm True if G is :

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Thm True if G is :

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③ Solvable

(Bartels-Lück-Reich 2008)

(Bartels-Lück 2009)

(Wegner 2013)



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$$\pi_i(\alpha_L) : \pi_i(BG_+ \wedge K_R^{GW}) \rightarrow K_i(R[G])$$

is an isomorphism for all $i \in \mathbb{Z}$.



My work is motivated by the Classical Farrell-Jones Conjecture.

*What Did I Do ?

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A formula for the composition
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Computing the
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Computing the
universal assembly
(if time permits)

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The diagram commutes :

$$\pi_1(BG_+ \wedge K_{\mathbb{Z}}^{GW})$$

$\cong$

$$K_1(\mathbb{Z}) \oplus [G_{ab} \otimes K_0(\mathbb{Z})] \quad (AHSS)$$

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$$\{\pm 1\} \oplus G_{ab}$$

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 K_1(\mathbb{Z}) \oplus [G_{ab} \otimes K_0(\mathbb{Z})] & & \\
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 \parallel & & \\
 \{\pm 1\} \oplus G_{ab} & \longrightarrow & K_1(\mathbb{Z}[G]) \\
 (\pm 1, [g]) & \longmapsto & \pm g
 \end{array}$$

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$i: R \hookrightarrow R[G]$ is the natural inclusion

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induced by

$$\begin{array}{ccc} G \times K_0(R) & \longrightarrow & K_1(R[G]) \\ (g, [P]) & \mapsto & \left\{ \begin{array}{l} R[G] \otimes P \longrightarrow R[G] \otimes P \\ u \otimes x \mapsto gu \otimes x \end{array} \right\} \end{array}$$

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My work generalises this.

For a regular ring R , the diagram commutes:

$$\pi_1(BG_+ \wedge K_R^{GW})$$

$\parallel$

$$K_1(R) \oplus [G_{ab} \otimes K_0(R)] \xrightarrow{K_1(i) \oplus \Phi_1} K_1(R[G])$$

induced by $G \times K_0(R) \longrightarrow K_1(R[G])$

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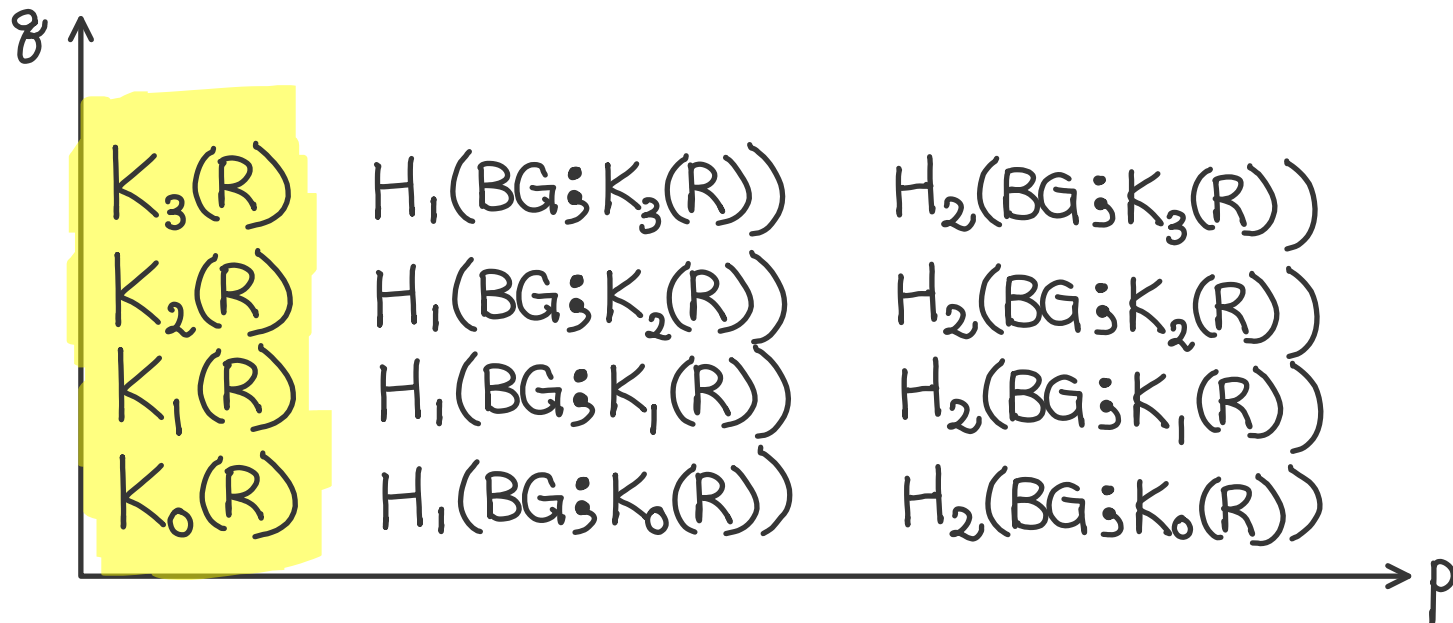
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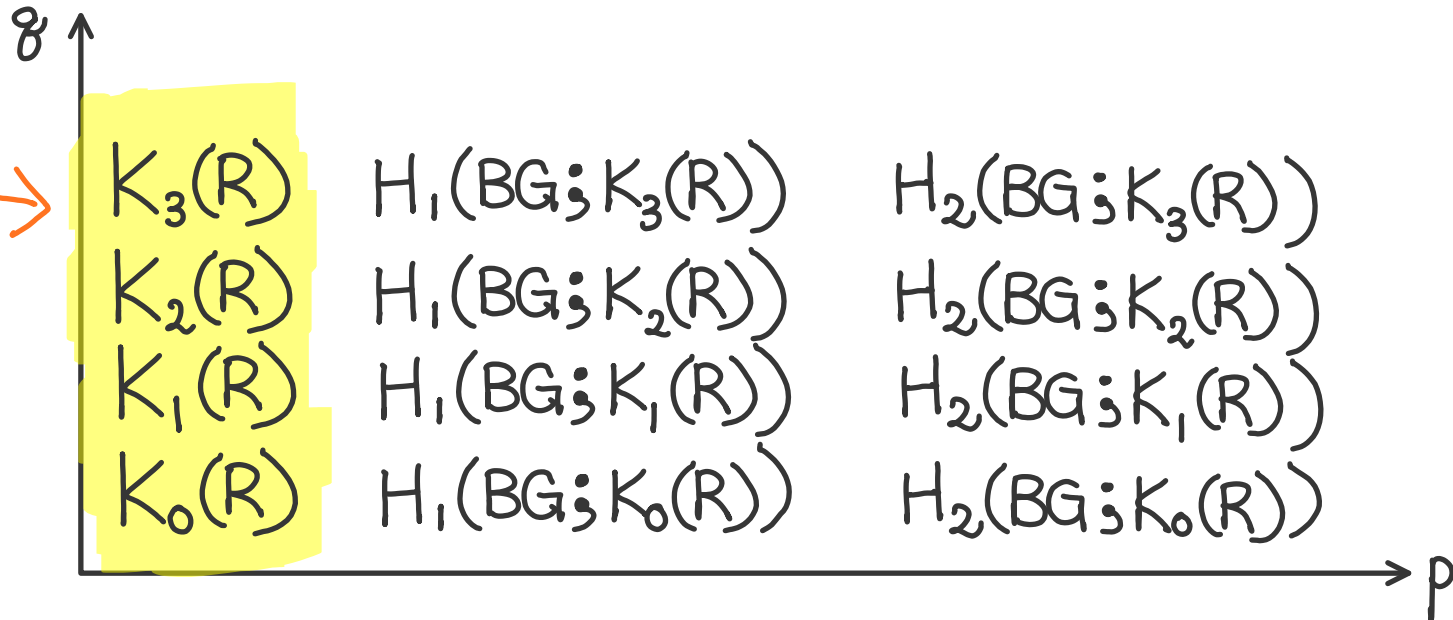
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$\pi_i(\alpha_L)$ is induced
by $R \hookrightarrow R[G]$

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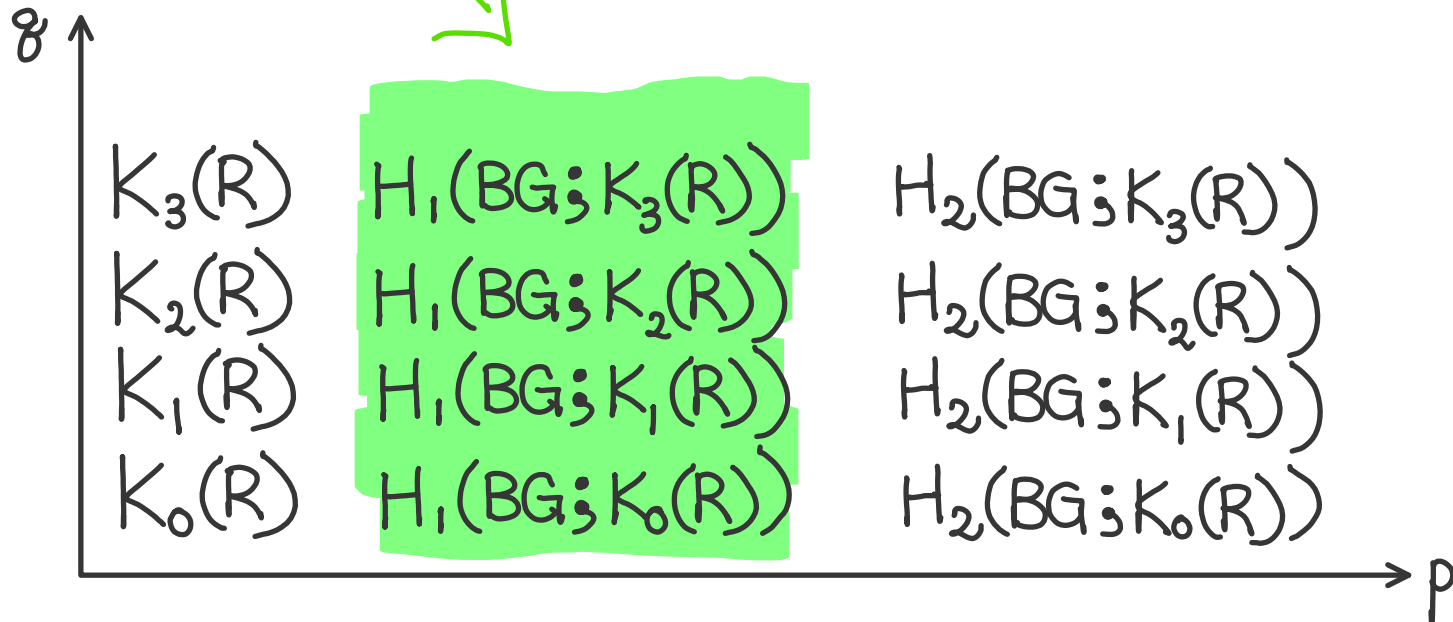
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My result

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$K_0(R)$	$H_1(BG; K_0(R))$	$H_2(BG; K_0(R))$

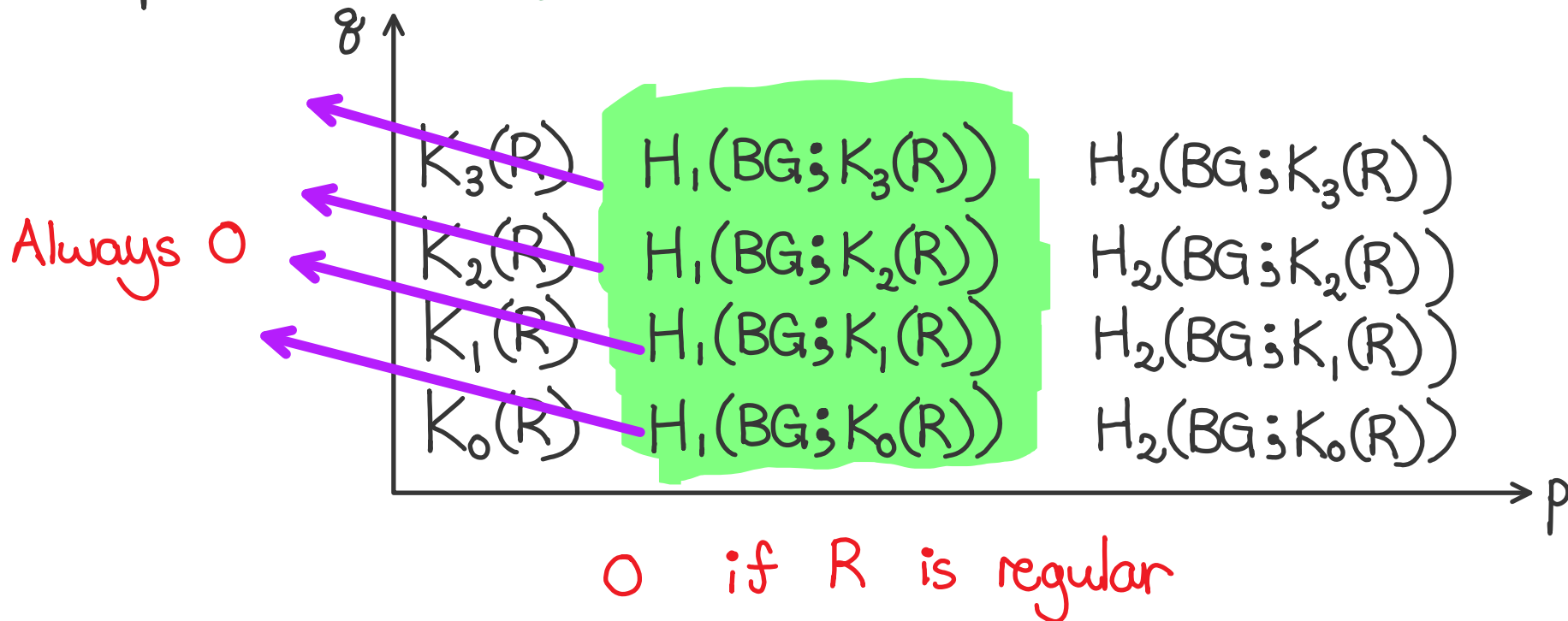
0 if R is regular

* The Formula

Recall the Atiyah-Hirzebruch spectral sequence

$$E_{p,q}^2 \cong H_p(BG; K_q(R)) \Rightarrow \pi_{p+q}(BG_+ \wedge K_R^{GW})$$

There is a subgroup $E_{1,i}^\infty$ of $\pi_{i+1}(BG_+ \wedge K_R^{GW})$, which is a quotient of $E_{1,i}^2 \cong H_1(BG; K_i(R))$

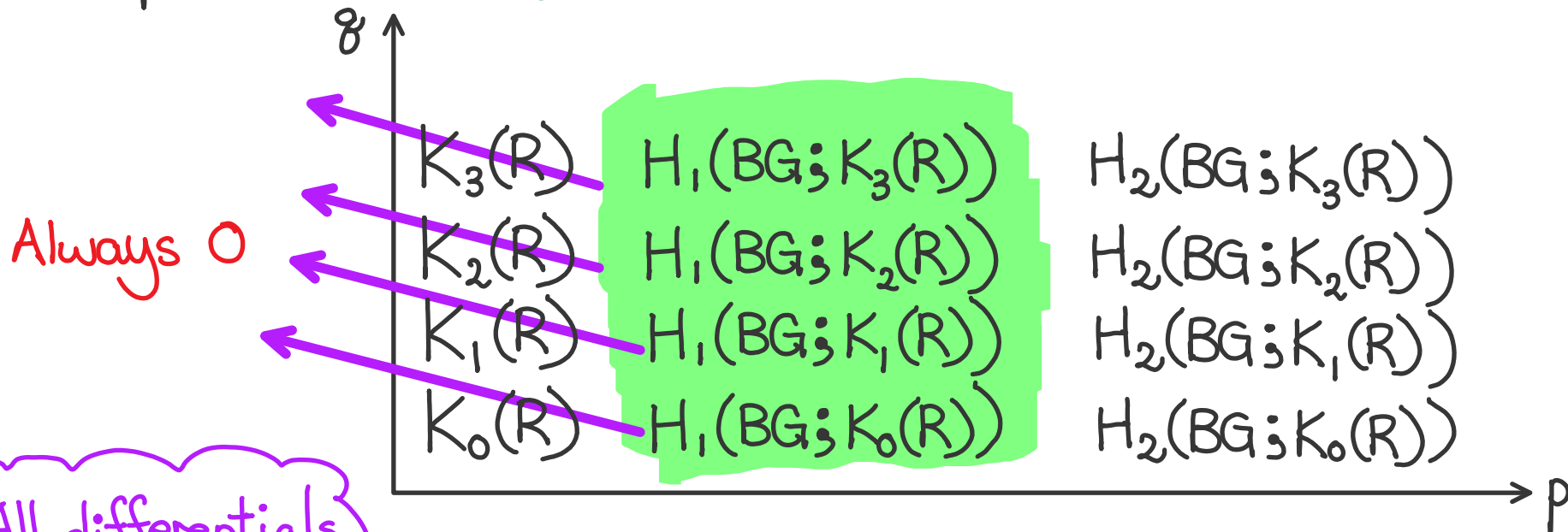


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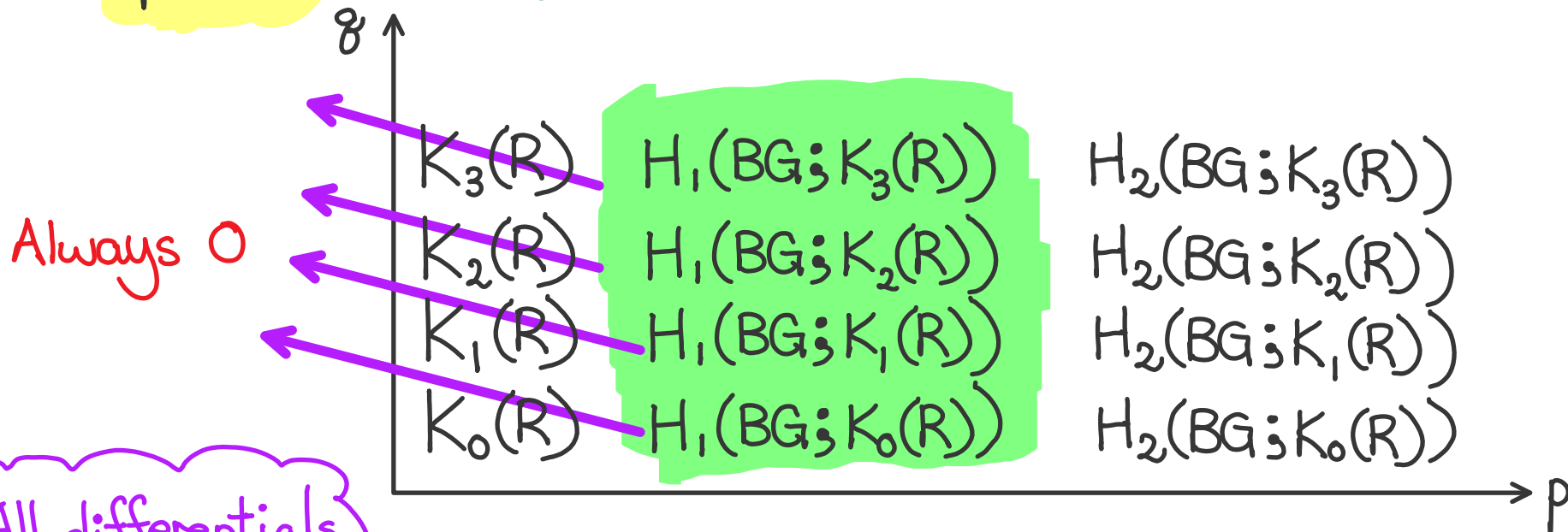
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* The Formula

$$E_{1,i}^2 \cong G_{ab} \otimes K_i(R)$$

$$\downarrow$$

$$E_{1,i}^\infty$$

$$\longrightarrow$$

$$K_{i+1}(R[G])$$

$$\uparrow \pi_{i+1}(\alpha_L)$$

$$\pi_{i+1}(BG_+ \wedge K_R^{GW})$$

* The Formula

Thm (C)

The filler

$$\begin{array}{ccc}
 E_{1,i}^2 \cong G_{ab} \otimes K_i(R) & \xrightarrow{\quad \overline{\Phi}_{i+1} \quad} & K_{i+1}(R[G]) \\
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where $\{g\} \in K_1(R[G])$ is represented by $g \in GL(1, R[G])$, and $\star'$ is the extended Loday product. ▣

* The Loday Product *

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{base-points}



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{ H-group structure
of $BGL(R \otimes S)^+$ }

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$$[\mathrm{K}_0(R) \times \mathrm{BGL}(R)^+] \wedge [\mathrm{K}_0(S) \times \mathrm{BGL}(S)^+] \xrightarrow{\gamma'} \mathrm{K}_0(R \otimes S) \times \mathrm{BGL}(R \otimes S)^+$$

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$$\mathbb{Z} \leftarrow \{ \text{Gersten-Wagoner delooping} \} \rightarrow \mathbb{Z}$$

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$$\left(\begin{aligned} (\$' \xrightarrow{f} \text{BGL}(\Sigma R)^+) \wedge (\$' \xrightarrow{g} \text{BGL}(\Sigma S)^+) &\longmapsto \left(\$' \wedge \$' \xrightarrow{f \wedge g} \text{BGL}(\Sigma R)^+ \wedge \text{BGL}(\Sigma S)^+ \right. \\ &\quad \left. \xrightarrow{\gamma} \text{BGL}(\Sigma^2(R \otimes S))^+ \right) \end{aligned} \right)$$

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$\wr$
 $\wr$

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well-defined up to weak homotopy

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If $[f] \in K_i(R)$ and $[g] \in K_j(S)$, then

$$[f] \star' [g] := [\chi' \circ (f \wedge g)] \in K_{i+j}(R \otimes S).$$

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But why not do

$$[K_0(R) \times BGL(R)^+] \wedge [K_0(S) \times BGL(S)^+] \xrightarrow{\gamma_{\text{naive}}} K_0(R \otimes S) \times BGL(R \otimes S)^+$$

$$([P], x) \wedge ([S], y) \mapsto ([P \otimes Q], \gamma(x, y))$$

instead?

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instead?

Because the associated multiplication

$$\star_{\text{naive}} : K_0(R) \otimes K_i(S) \rightarrow K_i(R \otimes S)$$

is zero for all $i > 0$.

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If $[P] \in K_0(R)$, $[Q] \in K_0(S)$, $\{u\} \in K_1(R)$ and $\{v\} \in K_1(S)$, then

$$\textcircled{1} \quad [P] \star' [Q] = [P \otimes Q]$$

$$\textcircled{2} \quad \{u\} \star' [Q] = \{u \otimes g + 1 \otimes (1-g)\}$$

$$\textcircled{3} \quad [P] \star' \{v\} = \{p \otimes v^{-1} + (1-p) \otimes 1\}$$

$$\textcircled{4} \quad \{u\} \star' \{v\} = \{u, v\}_{St} \quad (\text{Steinberg Symbol})$$

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Agree with the
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* The Proof

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$$\begin{array}{ccc}
 & g \otimes [f] & \longmapsto \{g\} \star' [f] \\
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Wolfgang Lück

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It suffices to prove
this for $G = \langle t \rangle$.



Wolfgang Lück

* The Proof

Prop When $G = \langle t \rangle$, the filler

$$\begin{array}{ccc}
 E_{1,i}^2 & \cong \langle t \rangle \otimes K_i(R) & \dashrightarrow K_{i+1}(R[t^{\pm 1}]) \\
 \downarrow & & \uparrow \pi_{i+1}(\alpha_L) \\
 E_{1,i}^\infty & \xrightarrow{\quad \quad \quad} & \pi_{i+1}(B\langle t \rangle_+ \wedge K_R^{GW})
 \end{array}$$

is induced by

$$\begin{aligned}
 \langle t \rangle \times K_i(R) &\longrightarrow K_{i+1}(R[t^{\pm 1}]) \\
 (t, [f]) &\longmapsto \{t\} \star' [f] .
 \end{aligned}$$



* The Proof

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$$\begin{aligned} G &\cong \text{Hom}(\langle t \rangle, G) \\ g &\mapsto (p_g: t \mapsto g) \end{aligned}$$

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 \downarrow & & \downarrow \\
 \pi_{i+1}(B\langle t \rangle \wedge K_R^{GW}) & \longrightarrow & \pi_{i+1}(BG \wedge K_R^{GW}) \\
 \pi_{i+1}(\alpha_L) \downarrow & & \downarrow \pi_{i+1}(\alpha_L) \\
 K_{i+1}(R[t^{\pm 1}]) & \longrightarrow & K_{i+1}(R[G])
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 \downarrow & & \downarrow \pi_{i+1}(\alpha_L) \\
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 \downarrow & & \\
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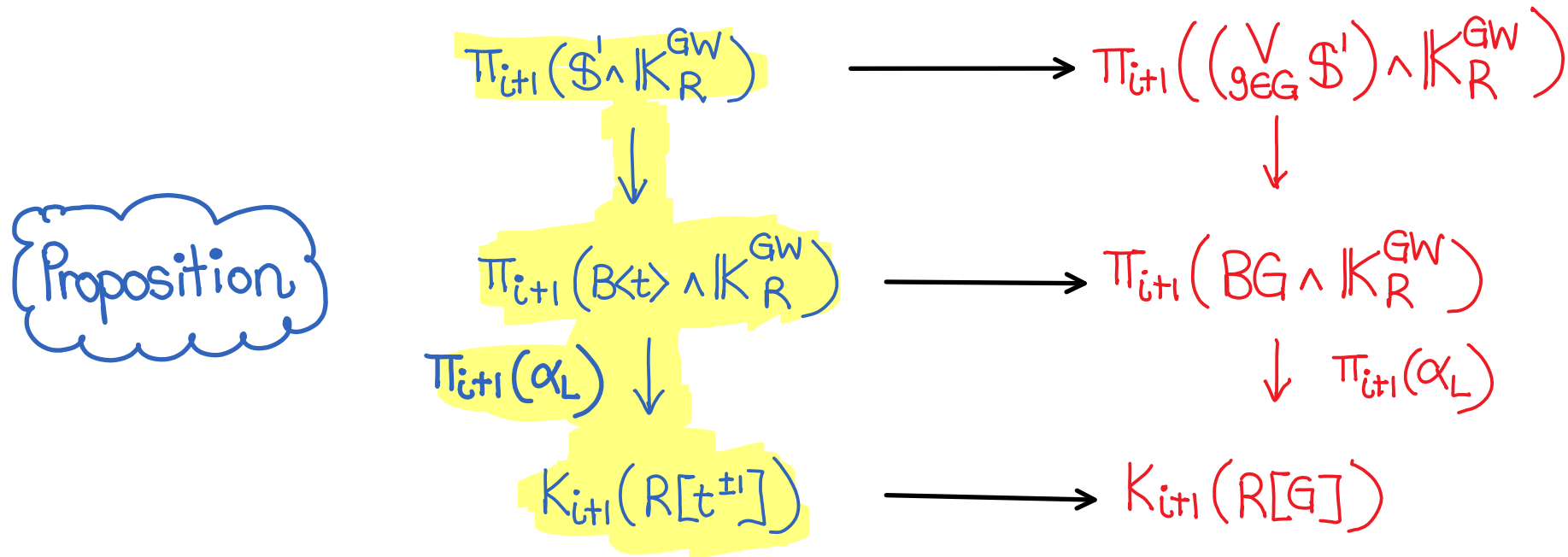
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which induces

$$\begin{array}{ccc} \pi_{i+1}(\mathcal{S}' \wedge K_R^{GW}) & \xrightarrow{(\varphi_g)_*} & \pi_{i+1}\left(\left(\bigvee_{g \in G} \mathcal{S}'\right) \wedge K_R^{GW}\right) \\ \downarrow & & \downarrow \\ \pi_{i+1}(B\langle t \rangle \wedge K_R^{GW}) & \xrightarrow{(\varphi_g)_*} & \pi_{i+1}(BG \wedge K_R^{GW}) \\ \pi_{i+1}(\alpha_L) \downarrow & & \downarrow \pi_{i+1}(\alpha_L) \\ K_{i+1}(R[t^{\pm 1}]) & \xrightarrow{(\varphi_g)_*} & K_{i+1}(R[G]) \end{array}$$

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For the general case, have

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 \downarrow & \swarrow \text{inclusion of 1-skeleton} \searrow & \downarrow \\
 \pi_{i+1}(B\langle t \rangle \wedge K_R^{GW}) & \xrightarrow{(\varphi_g)_*} & \pi_{i+1}(BG \wedge K_R^{GW}) \\
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$\begin{array}{ccc} K_i(R) & \xrightarrow{\varphi_g} & \pi_{i+1}((\bigvee_{g \in G} \mathcal{S}') \wedge K_R^{GW}) \\ \parallel & & \downarrow \\ \cong & & \pi_{i+1}(BG \wedge K_R^{GW}) \\ \downarrow & & \downarrow \\ \pi_{i+1}(\alpha_L) & & \pi_{i+1}(\alpha_L) \end{array}$

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* Corollary I : Formulas for π_1 & π_2

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$$\begin{array}{ccc}
 \textcircled{1} & \pi_1(BG_+ \wedge K_R^{GW}) & \xrightarrow{\pi_1(\alpha_L)} \\
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$$(g, [P]) \longmapsto \{g\} \star' [P] = \left\{ \begin{array}{l} R[G] \otimes P \longrightarrow R[G] \otimes P \\ u \otimes x \longmapsto gu \otimes x \end{array} \right\}$$

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② Have SES:

$$0 \rightarrow K_2(\mathbb{Z}) \oplus [G_{ab} \otimes K_1(\mathbb{Z})] \rightarrow \pi_2(BG_+ \wedge K_{\mathbb{Z}}^{GW}) \rightarrow H_2(BG; K_0(\mathbb{Z})) \rightarrow 0$$

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 & ([g], \pm 1) & \longmapsto & \{\pm 1, g\}_{St} & & &
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* Corollary II : Injectivity Problem for Π_2

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Is there exists a group G with torsions
and a regular ring R ,



James Davis

* Corollary II : Injectivity Problem for π_2

Is there exists a group G with torsions
and a regular ring R , so that
 $\pi_i(\alpha_L) : \pi_i(BG_+ \wedge K_R^{GW}) \rightarrow K_i(R[G])$
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2017

Then get non-injectivity on π_2

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What about when $R = \mathbb{Z}$?



James Davis



Mark Ullmann



Xiaolei Wu

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Note :

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Is there exists a group G with torsions
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Note : In this case, always have injectivity on π_0 and π_1 .

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So π_2 is the first place to look for non-injectivity!

* Corollary II : Injectivity Problem for Π_2

Thm (Joint with Ramras)

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Thm (Joint with Ramras)

The composition

$$K_2(\mathbb{Z}) \oplus [G_{ab} \otimes K_1(\mathbb{Z})] \hookrightarrow \pi_2(BG_+ \wedge K_{\mathbb{Z}}^{GW}) \xrightarrow{\pi_2(\alpha_L)} K_2(\mathbb{Z}[G])$$

$K_2(i) \oplus \overline{\Phi}_2$

----->

is always injective.



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Thm (Joint with Ramras)

The composition

$K_2(i) \oplus \bar{\mathbb{I}}_2$

Always split-injective

$$K_2(\mathbb{Z}) \oplus [G_{ab} \otimes K_1(\mathbb{Z})] \hookrightarrow \pi_2(BG_+ \wedge K_{\mathbb{Z}}^{GW}) \xrightarrow{\pi_2(\alpha_L)} K_2(\mathbb{Z}[G])$$

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The composition

{ Only need to check this }

$$K_2(i) \oplus \overline{\Phi}_2$$

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is always injective. ◻

◦◦ If $H_2(BG; \mathbb{Z}) = 0$, then $\pi_2(\alpha_L): \pi_2(BG_+ \wedge K_{\mathbb{Z}}^{GW}) \rightarrow K_2(\mathbb{Z}[G])$ is always injective.

* Corollary II : Injectivity Problem for π_2

$$G_{ab} \otimes K_1(\mathbb{Z}) \xrightarrow{\overline{\Phi}_2^G} K_2(\mathbb{Z}[G])$$

* Corollary II : Injectivity Problem for π_2

$$\begin{array}{ccc}
 G_{ab} \otimes K_1(\mathbb{Z}) & \xrightarrow{\overline{\Phi}_2^G} & K_2(\mathbb{Z}[G]) \\
 \tau \otimes \text{id} \downarrow & & \downarrow \nu_* \\
 V_{ab} \otimes K_1(\mathbb{Z}) & \xrightarrow{\overline{\Phi}_2^V} & K_2(\mathbb{Z}[V])
 \end{array}$$

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 \end{array}$$

$$\begin{array}{ccccc}
 \text{---} & \text{---} & \text{---} & \text{---} & \text{---} \\
 \nearrow & & \searrow & & \nearrow \\
 G & \twoheadrightarrow & G_{ab} & \longrightarrow & G_{ab} \otimes K_1(\mathbb{Z}) \\
 g & \mapsto & [g] & \mapsto & [g] \otimes (-1)
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Now, V is a vector space over $K_1(\mathbb{Z}) \cong C_2$. So, if $x \in V$ is non-zero, can find a projection

$$f_x : V \rightarrow C_2$$

$$x \mapsto \text{generator}$$

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 f_x \otimes \text{id} \downarrow & & \downarrow (f_x)_* \\
 C_2 \otimes K_1(\mathbb{Z}) & \xrightarrow{\overline{\Phi}_2^{C_2}} & K_2(\mathbb{Z}[C_2])
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$\parallel$

$\langle \{-1, -1\}_{\text{st}}, \{-1, 9\} \rangle$ (Dunwoody 1975)

$\parallel$

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generator of C_2

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 \end{array}$$

$$\begin{array}{c}
 \parallel \\
 \searrow \xrightarrow{g \otimes (-1)} \{ -1, g \} \hookrightarrow \langle \{ -1, -1 \}_{\text{st}}, \{ -1, g \} \rangle \quad (\text{Dunwoody 1975}) \\
 \parallel \\
 C_2 \oplus C_2
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 G_{ab} \otimes K_1(\mathbb{Z}) & \xrightarrow{\overline{\Phi}_2^G} & K_2(\mathbb{Z}[G]) \\
 \downarrow \tau \otimes \text{id} \quad \cong & & \downarrow \nu_* \\
 V \otimes K_1(\mathbb{Z}) & \xrightarrow{\overline{\Phi}_2^V} & K_2(\mathbb{Z}[V]) \\
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 \end{array}$$

$\xrightarrow{g \otimes (-1)} \{ -1, g \} \hookrightarrow \langle \{ -1, -1 \}_{\text{st}}, \{ -1, g \} \rangle \quad (\text{Dunwoody 1975})$

$\parallel$
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{ all are injective }

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Universal
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{ fundamental groupoid
of X }



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non-connective Pedersen-Weibel
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Thm (C)

For $i > 0$, the filler

$$E_{1,i}^2 \cong G_{ab} \otimes K_i(R) \xrightarrow{\hat{\Phi}_{i+1}} K_{i+1}(R[G])$$

$$\downarrow$$

$$E_{1,i}^\infty$$

$$\uparrow \pi_{i+1}(\hat{\alpha}_{BG})$$

$$\hookrightarrow$$

$$\pi_{i+1}(BG_+ \wedge K_R^{PW})$$

is induced by

$$G \times K_i(R) \longrightarrow K_{i+1}(R[G])$$

$$(g, [f]) \longmapsto \{g\} \star [f] \quad .$$



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The map $\hat{\Phi}_2 : G_{ab} \otimes K_1(\mathbb{Z}) \rightarrow K_2(\mathbb{Z}[G])$ is the same :
 $([g], \pm 1) \mapsto \{\pm 1, g\}_{st}$

So the composition

$$E_{1,1}^{\infty} \hookrightarrow \pi_2(BG_+ \wedge K_{\mathbb{Z}}^{PW}) \xrightarrow{\pi_2(\hat{\alpha}_{BG})} K_2(\mathbb{Z}[G])$$

is always injective.